

NS-3 Project Proposal

Base Paper

Title:

TCP Petra: Enhanced TCP Westwood Slow Start Phase

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Background Paper:

TCP Westwood: Bandwidth Estimation for Enhanced Transport over Wireless Links

Paper Link:

<https://ieeexplore.ieee.org/document/4448657>

(Background reference:

<https://web.cs.ucla.edu/classes/fall03/cs218/paper/p287-mascolo.pdf>)

Origin and Background

Traditional TCP congestion control mechanisms such as TCP Reno and NewReno rely heavily on packet loss as an indicator of congestion. In wireless networks, however, packet loss often occurs due to link errors rather than actual congestion, leading to poor performance.

TCP Westwood was proposed to address this issue by estimating available bandwidth using the rate of received ACKs and adjusting the congestion window accordingly. TCP Westwood+ further improves the stability of this estimation.

TCP Petra extends TCP Westwood+ by focusing on improving the **slow start phase**, which is often too aggressive in wireless environments and can cause congestion overshoot.

Problem Statement

Although TCP Petra improves slow start behavior compared to TCP Westwood+, it still relies on **fixed decision rules** for exiting the slow start phase. In highly dynamic wireless networks, round-trip time (RTT) can fluctuate significantly even before packet loss occurs.

As a result:

- Slow start may continue longer than optimal
- Congestion overshoot can still occur
- Delay increases are not sufficiently utilized as early congestion signals

There is a need for a **more adaptive slow start exit mechanism** that reacts to RTT dynamics instead of fixed thresholds.

Proposed Solution

This project proposes to extend TCP Petra by introducing **adaptive, RTT-based intelligence** into the slow start phase. The solution will be implemented and evaluated using the ns-3 network simulator.

Primary Modification (Main Contribution)

RTT-Variance-Aware Adaptive Slow Start Exit

- Continuously monitor RTT samples during slow start
- Compute RTT variance to capture delay instability
- Exit slow start early when RTT variance exceeds an adaptive threshold
- Transition smoothly into congestion avoidance

This approach aims to detect **incipient congestion** before packet loss occurs, especially in wireless networks with fluctuating delays.

Alternate Modification (Backup Option)

(To be used if the primary idea is already taken)

Hybrid Delay-Threshold-Based Slow Start Control

- Combine TCP Petra's logic with delay-based congestion detection
- Monitor minimum RTT (baseRTT) and current RTT
- Exit slow start if current RTT exceeds baseRTT by a predefined margin

This method introduces **delay awareness inspired by TCP Vegas**, but applies it specifically to the slow start phase of TCP Petra.

Implementation Plan

- Use ns-3's existing TCP Westwood+ implementation as the base
- Implement TCP Petra slow start logic
- Integrate the proposed adaptive modifications into the slow start phase
- Ensure minimal changes outside congestion control logic

Performance Evaluation Scope

The following variants will be compared:

- TCP Westwood+
- Original TCP Petra
- Modified TCP Petra (proposed)

Evaluation Metrics:

- Throughput
- RTT
- Packet loss rate
- Congestion window evolution

Scenarios:

- Wireless links with variable delay
- Error-prone wireless environments

Expected Contributions

- An adaptive enhancement to TCP Petra using RTT dynamics
- Improved slow start behavior under wireless conditions
- A clear demonstration of how delay variation can be used as an early congestion signal

Performance improvement is not the primary goal; the focus is on **understanding, correct implementation, and meaningful modification** of an existing protocol.

Conclusion

This project aims to enhance TCP Petra by introducing adaptive RTT-based slow start control mechanisms. By leveraging RTT variance and delay thresholds, the proposed modifications seek to reduce congestion overshoot and improve protocol responsiveness in wireless networks. The work aligns well with ns-3 capabilities and satisfies the requirement of independent modification and understanding.